The Influence of Facial Gender Characteristics on Time-Based Expectancy During a Gender Categorization Task

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Abstract

The aim of this bachelor's thesis was to examine the influence of facial gender characteristics on time-based expectancy during a gender categorization task.

The study used a version of the time–event correlation paradigm that we modified. Participants were shown female and male faces on a screen and had to identify the sex of the stimulus as quickly and accurately as possible. The stimuli were presented after a short or long foreperiod. In 90% of trials, a face of a particular sex was associated with a particular foreperiod, while an infrequent combination was presented in 10%.

Forty participants took part in the study, including 20 women and 20 men. Faces of young and older women and men were used, displaying neutral, positive, and negative emotional expressions. The data were analyzed in terms of reaction time.

The results showed that participants responded faster to predictable stimuli than to unpredictable stimuli, indicating an effect of time-based expectancy. In addition, an interaction between expectancy, stimulus sex, and participant sex was found: among female participants, the temporal expectancy effect emerged for female stimuli, whereas no such effect was found among male participants. Furthermore, no influence of the emotional expression or age of the stimuli on the formation of temporal expectancy was observed.

Keywords: time-based expectancy, gender categorization, gender bias, emotional expression, age.

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1. Introduction

The human social environment constantly demands fast and accurate responses. In everyday interactions, people often have very little time to decide whom they are interacting with, what emotional state the other person is in, and how to behave in a particular social situation.

Face perception is one of the important mechanisms for processing social information, because a human face provides the observer with information about a person's identity, emotional state, and intentions. Face perception differs from ordinary object recognition because faces form a relatively homogeneous visual category, and observers often need to distinguish small differences in facial features and their configuration. Contextual information can also influence face processing, indicating that faces are not always processed independently of the conditions in which they are perceived (Bruce & Young, 1986). The process of face perception is known to be implemented by a distributed neural system consisting of multiple bilateral regions. The representation of invariant aspects of faces underlies the recognition of individuals, while the representation of changeable aspects, such as gaze direction, expression, and lip movements, underlies the perception of information that facilitates social communication (Haxby et al., 2000).

Furthermore, when evaluating other people, observers often form and use categorical representations, such as sex, age, ethnicity, occupation, and so on, to simplify and accelerate person perception (Macrae & Bodenhausen, 2000). Thus, studying the factors that influence gender categorization is important not only in the context of visual recognition, but also as a model of the rapid processing of social information.

Emotional processes are known to serve not only to register the significance and value of sensory events, but also to elicit adaptive responses and alter perception. The emotional significance of a stimulus can influence sensory processing and attention. Experiments using behavioral methods, PET, and fMRI show that people attend more readily to emotional stimuli than to neutral stimuli (Vuilleumier, 2005).

Interestingly, age stereotypes are internalized throughout life, may operate unconsciously, become more significant when they become self-relevant, and act through several pathways.

Aging itself is partly a social construct (Levy, 2009). Accordingly, we can consider age as a category with social significance rather than a visual feature.

Time-based expectancy involves predicting when a particular event will occur. This process may not be conscious and changes dynamically in response to information from the environment, improving the processing of and responses to current sensory stimuli (Nobre et al., 2007). Since faces are one of the most important sources of information from the social environment, it is interesting to consider how temporal expectancy affects their processing. This issue has not been studied to date.

At the same time, social factors are not processed in isolation in everyday life. Accordingly, the aim of the present study is to examine together the roles of sex, age, emotional expression, and time-based expectancy in performance on a facial gender categorization task.

2. Literature Review

2.1. Face Perception and Gender

When discussing the categorization and perception of faces, it is important to consider whether people have certain types of biases. One such bias is gender bias, according to which face recognition may vary depending on whether the sex of the stimulus matches the observer's sex. Face perception is known to rely on a "face network" located in the ventral occipitotemporal cortex (Haxby et al., 2000, as cited in Lovén et al., 2014). Two areas are particularly important in this network: the inferior occipital gyrus (IOG), which is involved in the early visual analysis of facial features, and the fusiform gyrus (FFG), which is associated with processing facial features and their configuration (Haxby et al., 2000; Liu et al., 2010, as cited in Lovén et al., 2014). This means that processing facial sex is a multilevel process that may begin with early visual analysis and continue with higher-level configural processing.

Studies of own-group bias show that people often process faces of members of their own group more efficiently than faces of members of other groups. For example, viewing own-race faces often produces a higher BOLD response in the fusiform gyrus than viewing other-race faces (Golby et al., 2001; Feng et al., 2011, as cited in Lovén et al., 2014). The BOLD response is a

signal measured with fMRI that reflects changes in blood oxygen levels and is associated with activity in a particular brain area. In other words, a higher BOLD response indicates that this area was more strongly engaged in processing a particular face. This suggests that social categories, such as race or sex, may alter face processing at the neural level (Golby et al., 2001; Feng et al., 2011, as cited in Lovén et al., 2014). An increased BOLD response to own-group faces in the IOG also suggests that the influence of such social categories may emerge at an early visual stage of face processing (Feng et al., 2011, as cited in Lovén et al., 2014).

An fMRI study conducted by Lovén and colleagues in 2014 showed that facial sex influences neural activity associated with face processing. At the behavioral level, female participants recognized female faces better than male faces. No such difference was found in men: they recognized female and male faces approximately equally well. This suggests that women may exhibit an own-sex bias in face recognition (Lovén et al., 2014).

In the same study, the FFG was more strongly activated in women when they viewed female faces than when they viewed male faces. In men, facial sex did not have such a clear influence on FFG activity. This shows that facial sex may also be processed differently at the brain level, particularly in women (Lovén et al., 2014). The IOG in women was also more strongly activated by female faces than by male faces. This is important because it suggests that the influence of facial sex may not be limited to later cognitive evaluation and may also be apparent at an early stage of face processing (Lovén et al., 2014).

This study showed that facial sex is a socially significant category that may influence not only behavioral outcomes but also brain activity (Lovén et al., 2014). Therefore, in a gender categorization task, a participant does not simply distinguish “female” from “male,” but processes a social stimulus whose processing may be influenced by the observer’s sex, experience, attention, and attitude toward the social category (Lovén et al., 2014).

In this respect, a study conducted by Hausinger and colleagues in 2025 is of interest. It showed that women’s own-gender bias may not be explained solely by better recognition of whole faces. Women in the luteal phase were particularly accurate in recognizing individual parts of female faces: the eyes, nose, or mouth. This suggests that women’s better recognition of female faces

may be based not only on better processing of the whole face, but also on more detailed representations of facial parts (Hausinger et al., 2025).

In the study, holistic face processing was assessed with a part–whole task. In this task, participants recognize facial parts both in the context of a whole face and in isolation. If a facial part is recognized more accurately in the context of a whole face, this indicates holistic processing. The study found a clear part–whole effect among male participants, indicating that they recognized facial parts more accurately when those parts were presented in the context of a whole face. Among female participants in the luteal phase, no overall part–whole effect was found, although facial gender played an important role: the part–whole effect was present for female faces, but not for male faces. This suggests that, among women in the luteal phase, holistic processing was particularly associated with own-sex faces. At the same time, their advantage in recognizing female faces was especially apparent in the isolated facial-parts condition as well, showing that women's own-gender bias may be based on more detailed processing of facial parts as well as whole faces (Hausinger et al., 2025).

This study shows that the processing of facial gender may be influenced not only by the sex of the stimulus, but also by the observer's biological or hormonal state (Hausinger et al., 2025).

Previous research on gender differences in face recognition mainly concerned the processing of unfamiliar faces. Such studies often found a tendency for women to recognize own-gender faces better. A study conducted by Mishra and colleagues in 2019 examined this issue in the context of familiar/celebrity face recognition. The authors note that unfamiliar faces are processed less efficiently than familiar faces because familiar faces have more stable, invariant representations (Mishra et al., 2019).

Their results showed no significant difference between male and female participants in overall celebrity face recognition accuracy. This differs from studies of unfamiliar faces, in which women often perform better. Accordingly, the authors suggest that sufficient experience and knowledge may reduce or eliminate the gender differences observed in unfamiliar face recognition (Mishra et al., 2019).

Notably, in this study, an own-gender bias for celebrity faces was found mainly among male participants rather than women. This finding differs from research on unfamiliar faces, where own-gender bias is more frequently reported in women. In addition, across all countries, male celebrities' faces were generally recognized more accurately than female celebrities' faces. The authors suggest that better recognition of male celebrities' faces may be explained by the fact that male celebrities appear more frequently in the media. Participants therefore had greater exposure to their faces, which facilitated recognition (Mishra et al., 2019).

The role of sociocultural gender equality also proved important. In countries with low gender equality, female participants remembered female celebrities' faces better than male participants did, whereas this difference was no longer observed in countries with high gender equality. This suggests that gender differences in face recognition may depend not only on individual or biological factors, but also on the sociocultural context (Mishra et al., 2019).

Thus, face recognition is a complex process that depends on many factors, including gender bias.

2.2. Gender Categorization and Emotional Expression

Social perception is not simply the passive reception of information; it helps people decide which social objects to approach and which to avoid. In this context, facial expression is an important social signal: a smile may be perceived as a signal of approach and positive interaction, whereas anger may signal threat and avoidance. Such signals may also be associated with gender cues: anger may be more strongly associated with masculine facial features, and happiness with feminine facial features (Becker et al., 2007).

In general, stimuli associated with negative and unpleasant objects are identified more slowly than those associated with positive and pleasant objects (Fazio, 2001, as cited in Becker et al., 2007). One explanation for this slowing is that negative stimuli automatically activate subcortical alertness and threat-monitoring mechanisms, which delay the classification of these stimuli by directing cognitive resources toward other, defensive processes (Becker et al., 2007). Research has also found that participants needed more time to count the features of schematic faces when

the faces displayed negative expressions than when they displayed positive expressions (Eastwood et al., 2003, as cited in Becker et al., 2007).

A study conducted by Becker and colleagues (2007) showed that participants were more likely to imagine a man when imagining an angry face, and a woman when imagining a happy face. Furthermore, anger was recognized faster and more accurately on male faces, whereas happiness was recognized faster and more accurately on female faces. Interestingly, angry faces were categorized faster and more accurately as male faces, and happy faces as female faces (Becker et al., 2007).

Thus, emotional expression may also influence gender categorization. When studying this issue, emotional expression should also be taken into account, particularly when the stimuli include happy, angry, and neutral faces.

2.3. The Role of Age in Face Perception

In addition to the relationship between gender and emotional expression, it is important to consider another social characteristic of faces: age.

Own-age bias is an important phenomenon in face perception and refers to people's better recognition of faces from their own age group than faces from other age groups (Cronin et al., 2019).

The authors discuss two main approaches to explaining own-age bias. The first is the perceptual-expertise account, according to which people recognize faces from the age group with which they have more frequent contact better. Because people often spend more time with others of their own age, they have more experience distinguishing and encoding these faces. The second is the social-cognitive account, according to which face memory is influenced by how we evaluate a particular person or group. In-group faces may be processed more individually, whereas out-group faces may be processed more categorically, making their later recognition more difficult (Cronin et al., 2019).

Their results showed that emotional expression reduced own-age bias mainly by improving the recognition of older faces. In other words, when emotional expressions were present, faces from another age group were processed and recognized better than when the expressions were

neutral. According to the authors, this suggests that own-age bias cannot be explained solely by perceptual expertise and also involves social-cognitive factors (Cronin et al., 2019).

In addition to own-age bias, a face may be perceived differently depending on the visual context in which it is presented. For example, the age of surrounding faces has been shown to influence judgments of a target face's age. When the target face was accompanied by young faces, participants judged it to be younger (Pilz & Lou, 2022).

The authors relate this finding to a perceptual averaging mechanism: when several faces are presented together, the observer's perception may shift partly toward the average characteristics of those faces (Pilz & Lou, 2022).

These studies show that facial age and emotional expression should not be considered independently of one another, and support the view that sex, age, and emotional expression are interconnected social cues in face perception (Cronin et al., 2019). Furthermore, facial age is not perceived in isolation and is influenced by the visual context.

2.4. Time-Based Event Expectancy and the Socioemotional Context

The studies discussed above show that face perception depends not only on the physical characteristics of a stimulus, but also on context, emotional expression, and the social categories used by the observer. However, face processing may be influenced by temporal as well as visual context. It is therefore important to consider the mechanisms of time-based expectancy and their role in perception and response preparation.

Time-based expectancy is associated with predicting that a particular event will occur at a particular time. Such expectancy does not necessarily involve conscious awareness. It is continuously formed and updated, and helps people respond better to ongoing sensory stimulation (Nobre et al., 2007).

Furthermore, when a stimulus appears at a predictable time, participants can prepare their responses better, whereas such preparation is more difficult under unpredictable conditions. Research has demonstrated that temporal predictability reduces reaction time and improves behavioral responses (Nobre et al., 2007).

The influence of temporal prediction is not limited to motor responses. Temporal certainty between events may alter perceptual thresholds and increase the discriminability of visual stimuli. Perceptual judgments improve when a stimulus appears within a regular rhythm and at an expected time, whereas accuracy decreases when a stimulus is presented earlier or later than expected. This suggests that temporal expectancy may affect both response preparation and the perceptual processing of information (Nobre et al., 2007).

Specific temporal expectancy can be defined as the expectation of a combination of an event and a particular point in time. The event is not expected in itself, but in association with a specific time. Specific temporal expectancy is widespread in all domains of human behavior in which the duration of an interval predicts an event. For example, in spoken language, pause duration influences listeners' expectations about the complexity of the expression that follows. Similarly, in human–computer interaction, the duration of a delay often predicts the computer's response, and users adapt to these regularities. For example, after navigating to a webpage, a person's expectation that it will load successfully gradually decreases as time passes, eventually changing into an expectation of an error message. Research confirms that stimuli are processed faster and more accurately in predictable cases (Thomaschke & Dreisbach, 2013).

It is also important that various factors influence the formation of temporal expectancy, one of which is emotion.

Positive mood has been found to increase attentional prioritization of information that is closer in time. The formation of time-based expectancy was observed for the short foreperiod in the positive-mood group, and for the long foreperiod in the neutral-mood condition (Kunchulia et al., 2021).

This study showed that time-based expectancy is not determined solely by the temporal predictability of a stimulus and may also be influenced by a person's emotional state. According to the authors, positive mood increases attentional prioritization of information expected closer in time. The study confirms that temporal expectancy is not an isolated cognitive mechanism and may interact with emotional context (Kunchulia et al., 2021).

Interestingly, self-related cognition modulates temporal expectancy. For example, in a task-switching paradigm, the time-based expectancy effect has been shown to emerge only for the neutral task: participants responded faster in trials in which the interval predicted the neutral task type than in temporally unpredictable neutral trials. However, no such effect emerged for the self-related task. This means that time-based event expectancy was restricted to the neutral task (Kunchulia et al., 2022).

These studies again confirm that social and cognitive factors may alter the formation of temporal expectancy.

2.5. Novelty of the Study

Based on the studies discussed above, we can therefore conclude that a gender categorization task may be influenced by many factors, such as age, participants' subjective attitudes, emotional state, and context. Nevertheless, these factors have not been studied in combination. Within the scope of this bachelor's thesis, the influence of time-based expectancy on gender categorization was examined for the first time, along with the extent to which this process is affected by gender bias, age, emotional expression, and the interactions among these factors. For this purpose, a modified version of the classical time–event correlation paradigm was developed. In the classical paradigm, two target stimuli and two preparatory intervals, or foreperiods, occur with equal frequency, but in most trials one stimulus is paired with the short foreperiod and the other with the long foreperiod (Wagener & Hoffmann, 2010). In the present study, this principle was applied to a gender categorization task: in 90% of cases, stimulus sex was paired with a short or long foreperiod, while an infrequent combination was presented in 10% of cases. The paradigm was counterbalanced across participants: for 50% of participants, the short foreperiod was more frequently associated with a male face and the long foreperiod with a female face; for the other half, the reverse was true.

Several main hypotheses were formulated for the study:

- According to the first hypothesis, temporal expectancy would improve performance on the facial gender categorization task. In this bachelor's study, comparing predictable and unpredictable conditions made it possible to assess how temporal expectancy affects the

speed of gender categorization. Specifically, predictable stimuli would be processed faster and more accurately than unpredictable stimuli.

- According to the second hypothesis, sex biases would influence the formation of temporal expectancy. By comparing the two gender groups of participants, male and female, we expected temporal expectancy to be more pronounced for female face stimuli among women and, conversely, for male face stimuli among men.
- According to the third hypothesis, emotional and social factors (emotion and age) and time-based expectancy would interact with one another during performance of the gender categorization task.

3. Methodology

3.1. Participants

Forty participants took part in the study. Their ages ranged from 17 to 23 years (M_age = 19.6, SD = 1.5), and they included 20 females and 20 males. Three participants were left-handed and the remainder were right-handed; all participants completed the experiment using their dominant hand.

Before the experiment, each participant completed a general information questionnaire through which we obtained information about their age, sex, and physical and mental health. No congenital disease or mental disorder was identified in any participant, and none was taking any medication affecting cognitive functions at the time of the experiment. A sample general information form is provided in the appendix (Appendix 1).

The experiment was conducted in accordance with the World Medical Association's Declaration of Helsinki. Each participant provided written consent to take part in the study. A sample consent form is provided in the appendix (Appendix 2).

3.2. Questionnaires

Satisfaction With Life Scale: Before the experiment, all participants completed the Georgian-language Satisfaction With Life Scale, which had been translated from English for this study. The standard back-translation procedure was followed (Brislin, 1970): first, the scale was

translated from English into Georgian, then from Georgian into English, after which the final version was reconciled (Diener et al., 1985). Based on this questionnaire, we determined participants' overall levels of subjective well-being and satisfaction with life. The questionnaire consists of five statements to which participants respond using a 7-point scale, where a higher score indicates greater life satisfaction and a lower score indicates the reverse. We used the questionnaire to assess participants' general psychological state and to account for subjective differences in their well-being (Appendix 3).

"In most ways, my life is close to my ideal."

"The conditions of my life are excellent."

"I am satisfied with my life."

Anger Scale of the Aggression Questionnaire: The study used the Anger Scale of the Aggression Questionnaire (Buss & Perry, 1992) to determine participants' levels of proneness to anger and irritation. Back-translation was also carried out in this case. This scale consists of seven statements assessing an individual's irritability, the intensity of affective reactions, and difficulty controlling anger. Higher scores indicated higher levels of anger (Appendix 4).

"I get angry easily, but the feeling soon passes."

"When I am frustrated, I show my irritation."

"Sometimes I feel like a powder keg ready to explode."

"I am an even-tempered person." (reverse-scored)

Questionnaire for Assessing Negative Stereotypes About Older People: A questionnaire assessing negative stereotypes about older people was also used (Montorio & Izal, 1991, as cited in Bahillo-Ruiz et al., 2025) to determine participants' levels of age stereotypes and prejudices toward older people. Back-translation was also used when translating this questionnaire. It consists of 20 negatively worded statements concerning attitudes toward health, intellectual competence, social functioning, activity, personality characteristics, and ability to work. The questionnaire was important for identifying participants' age stereotypes and considering the possible relationship between this variable and the experimental results. A lower

total score indicated a higher level of ageism, whereas a higher score indicated the reverse (Appendix 5).

“Most adults maintain a satisfactory level of health until approximately age 65, after which their health deteriorates sharply.”

“Most people over 65 have a range of limitations that make them dependent on others.”

“Most people begin to experience a significant decline in memory at around age 65.”

For all questionnaires, a total score was calculated from participants’ responses. Scores on the Satisfaction With Life Scale and the questionnaire assessing negative stereotypes about older people were summed in the standard way, while one statement in the Anger Scale of the Aggression Questionnaire was reverse-scored. Participants’ levels of life satisfaction, anger, and ageism were assessed based on the resulting total scores.

Table 1. Questionnaires used in the study

Variable	N	Mean (M)	Standard deviation (SD)
Life satisfaction (Happiness)	40	20.12	5.89
Negative stereotypes about older people (Ageism Scale)	40	49.95	5.71
Anger Scale (Aggression)	40	18.17	6

3.3. Apparatus and Stimuli

To conduct the experiment and collect data, we used E-Prime 2 software (Schneider, Eschman, & Zuccolotto, 2002) and the following computer model: a Windows PC with an LCD display (screen resolution 1280 × 800 pixels).

Four basic facial identities were used in the experiment: a young man, an older man, a young woman, and an older woman. Each was presented with three different emotional expressions: neutral, positive, and negative. Thus, a total of 12 visual stimuli were used in the experiment. The fixation point was a black “+” symbol. All of these stimuli appeared in the center of the computer screen.

The stimuli were selected from a publicly available online database (https://faces.mpdl.mpg.de/imeji/) containing photographs of emotional facial expressions of people of different ages and sexes (Ebner et al., 2018). The intertrial interval (ITI) was 1000 ms (see Figure 1).

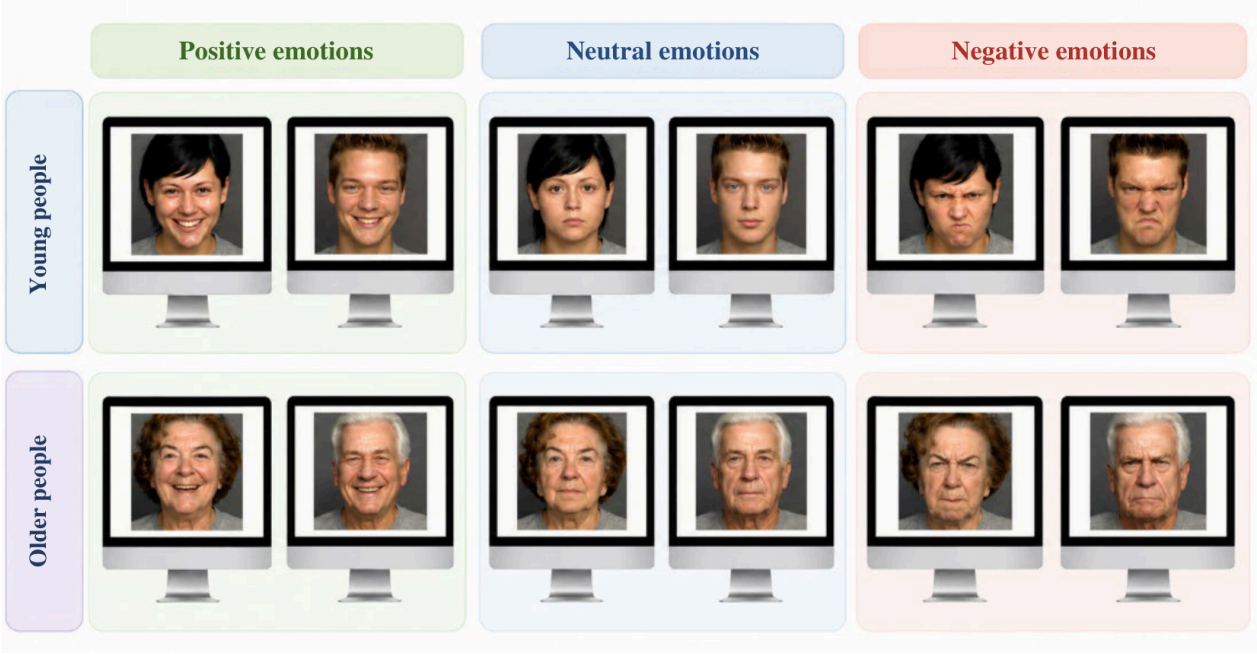


Figure 1. Emotional facial stimuli used in the experiment

3.4. Experimental Procedure

The experiment was conducted in the laboratory of the Institute of Cognitive Neuroscience at the Free University of Tbilisi. After completing the questionnaires, participants performed the time-based sex categorization paradigm that we developed for this study to investigate the interaction among temporal expectancy, age, and emotion.

Participants performed the task according to the following instructions, which they received before the test: “Female and male faces will appear randomly on the screen. If a female face appears, press the marked right-hand key on the keyboard, and if a male face appears, press the left-hand key. Please perform the task as quickly and accurately as possible.”

Participants’ task performance was assessed by reaction time in the female and male face identification task.

The procedure was identical for all participants. Stimuli were presented randomly. The experiment was divided into three blocks, between which participants could rest if they wished and continue the test after a short interval. The total duration of the experiment was 40–50 minutes.

The experiment comprised 720 trials in total. In each trial, a warning symbol (+ sign) appeared on the screen and remained visible for different intervals during the so-called foreperiod: 600 or 1200 ms. This was followed by a stimulus, a human face that was male or female, older or younger, with a negative, positive, or neutral emotional expression. Participants had to distinguish its sex by pressing the right or left key.

In 90% of cases, stimulus sex was paired with a short or long foreperiod, whereas in 10% of cases there was an infrequent combination of foreperiod and stimulus. Furthermore, for 50% of participants, the short foreperiod was paired with male faces and the long foreperiod with female faces, while the reverse was true for the other half (see Figure 2).

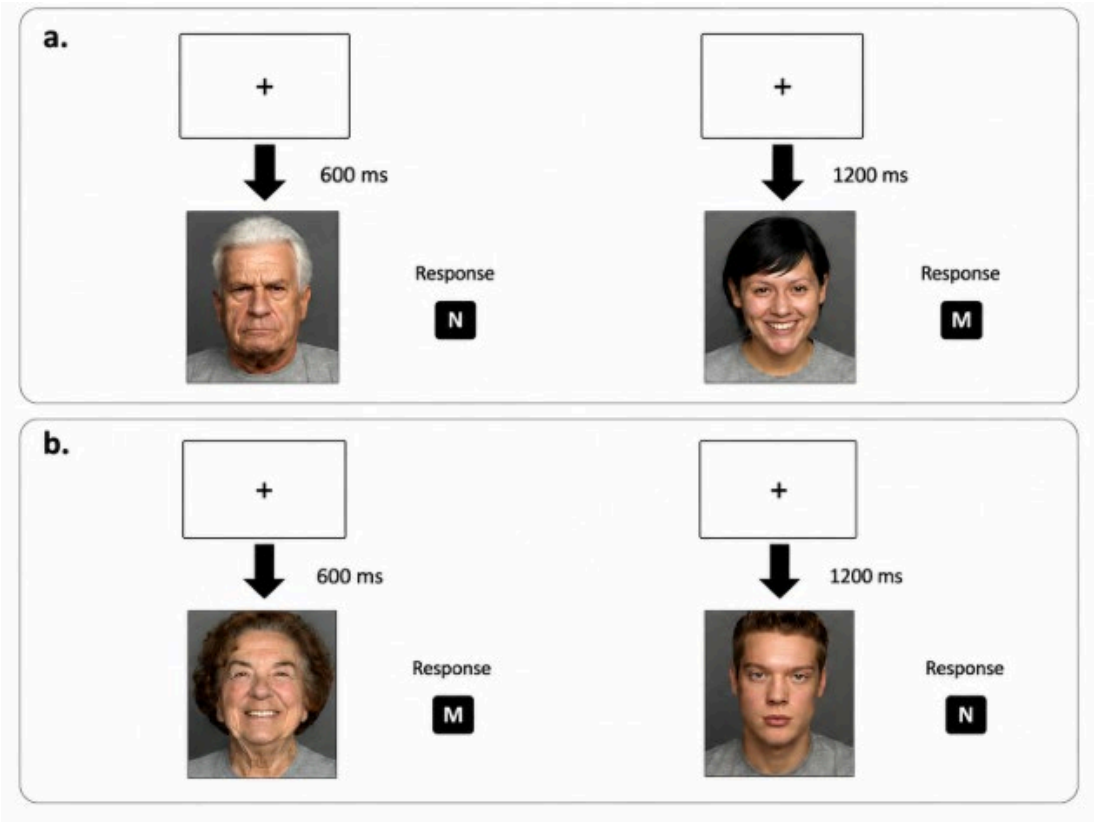


Figure 2. Schematic illustration of the task: a. The short foreperiod is paired with male faces and the long foreperiod with female faces. b. The short foreperiod is paired with female faces and the long foreperiod with male faces.

3.5. Data Processing

At the first stage of data analysis, the first 20 of the 720 trials for each participant were treated as practice trials. Accordingly, the first 20 trials of each participant were not used in the final data analysis.

Trials in which participants gave an incorrect response were not included in the reaction-time analysis. Thus, reaction times were analyzed only for correct responses, to assess task performance speed in cases where participants correctly identified the sex of the stimulus. Responses with reaction times below 200 ms or above 2000 ms were also excluded from the analysis. In addition, responses whose reaction times exceeded three standard deviations from the mean were not included.

A mixed ANOVA was used for statistical analysis, with four variables defined as within-subject factors. The first factor was the emotional expression of the stimulus, comprising three categories: neutral, positive, and negative. The second factor was stimulus age, comprising two categories: young and older. The third factor was expectancy/frequency, comprising frequent and infrequent combinations. The fourth factor was stimulus sex, consisting of female and male faces. The between-subject factor was the female and male group.

4. Results

The mixed ANOVA of reaction time revealed a main effect of expectancy, $F(1,38) = 4.38$, $p = 0.043$, indicating that participants responded faster to predictable stimuli than to unpredictable stimuli. A statistically significant interaction was also found among expectancy, stimulus sex, and group, $F(1,38) = 4.28$, $p = 0.046$. The remaining effects and interactions were not statistically significant (see Table 2).

Table 2. Mixed ANOVA data:

Effect	df	F	p	Partial Eta Squared
Expectancy	1.38	4.379	.043	.103
Expectancy × Group	1.38	1.760	.193	.044
Gender	1.38	.017	.898	.000
Gender × Group	1.38	1.155	.289	.029
Stimulus emotion	2.76	.346	.708	.009
Emotion × Group	2.76	.715	.492	.018
Stimulus age	1.38	1.101	.301	.028
Age × Group	1.38	.408	.527	.011
Expectancy × Gender	1.38	1.383	.247	.035
Expectancy × Gender × Group	1.38	4.276	.046	.101
Expectancy × Stimulus emotion	2.76	.418	.660	.011
Expectancy × Emotion × Group	2.76	.623	.539	.016

Effect	df	F	p	Partial Eta Squared
Gender × Emotion	2.76	1.756	.180	.044
Gender × Emotion × Group	2.76	.555	.577	.014
Expectancy × Gender × Emotion	2.76	1.570	.215	.040
Expectancy × Gender × Emotion × Group	2.76	.200	.819	.005
Expectancy × Age	1.38	.124	.726	.003
Expectancy × Age × Group	1.38	.032	.860	.001
Gender × Age	1.38	.201	.657	.005
Gender × Age × Group	1.38	.847	.363	.022
Expectancy × Gender × Age	1.38	.222	.640	.006
Expectancy × Gender × Age × Group	1.38	.167	.685	.004
Emotion × Age	2.76	.782	.461	.020
Emotion × Age × Group	2.76	.694	.503	.018
Expectancy × Emotion × Age	2.76	.121	.886	.003
Expectancy × Emotion × Age × Group	2.76	.806	.451	.021
Gender × Emotion × Age	2.76	.202	.818	.005
Gender × Emotion × Age × Group	2.76	1.377	.258	.035
Expectancy × Gender × Emotion × Age	2.76	.155	.856	.004
Expectancy × Gender × Emotion × Age × Group	2.76	1.050	.355	.027

To clarify the statistically significant interaction among expectancy, stimulus sex, and group, paired t-tests were additionally conducted for each group (women and men). These revealed that the interaction was due to female participants, unlike male participants, forming temporal expectancy only for own-sex stimuli, $t(19) = -2.955$, $p = 0.008$ (see Tables 3 and 5). They did not form temporal expectancy for male face stimuli, $t(19) = .463$, $p = .649$ (see Tables 3 and 5; Figure 3). Men did not exhibit an own-gender bias, that is, for male face stimuli, $t(19) = -1.070$, $p = .298$ (see Tables 4 and 6; Figure 3).

Table 3. Paired t-test results for the female group

Pair		t	Df	p
Pair 1	Male face_frequent combination Male face_infrequent combination	.463	19	.649
Pair 2	Female face_frequent combination Female face_infrequent combination	-2.955	19	.008

Table 4. Paired t-test results for the male group

Pair	Comparison	t	df	p
Pair 1	Male face_frequent combination Male face_infrequent combination	-1.070	19	.298
Pair 2	Female face_frequent combination Female face_infrequent combination	-.034	19	.973

Table 5. Mean reaction times of the female group in frequent and infrequent combinations

Variable	Mean	N	Standard deviation	Standard error of the mean
Male face_frequent combination	602.3186	20	102.81477	22.99008
Male face_infrequent combination	598.0708	20	99.64825	22.28203
Female face_frequent combination	593.3466	20	100.31666	22.43149
Female face_infrequent combination	618.6473	20	114.27342	25.55231

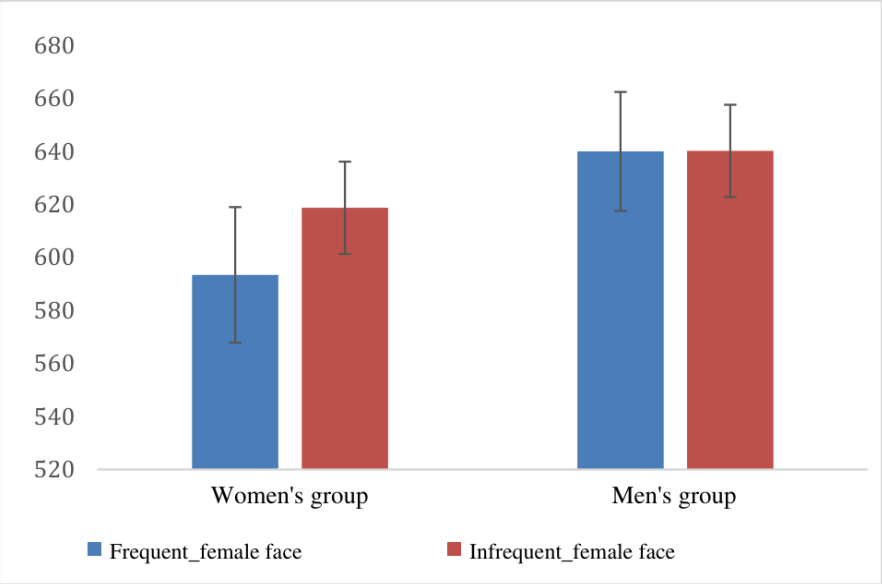
Table 6. Mean reaction times of the male group in frequent and infrequent combinations

Variable	Mean	N	Standard deviation	Standard error of the mean
Male face_frequent combination	640.5135	20	72.54590	16.22176
Male face_infrequent combination	646.1755	20	78.64236	17.58497

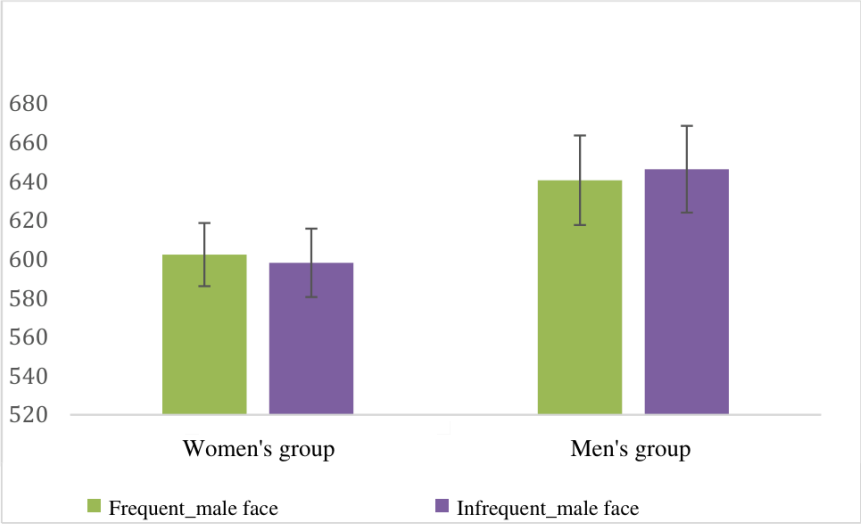
Variable	Mean	N	Standard deviation	Standard error of the mean
Female face_frequent combination	639.8798	20	77.76469	17.38871
Female face_infrequent combination	640.0724	20	77.82405	17.40199

Figure 3. Participants' mean reaction times by time-based expectancy and stimulus sex. a. For female face stimuli. b. For male face stimuli.

a.



b.



To determine whether individual differences and personality characteristics were related to the size of the temporal expectancy effect, an exploratory correlational analysis was conducted. Pearson correlations were calculated between expectancy magnitude (reaction time for the infrequent combination minus that for the frequent combination) and scores on the questionnaire assessing negative stereotypes about older people, the Satisfaction With Life Scale, and the Anger Scale of the Aggression Questionnaire. Correlations among the variables examined were not statistically significant (see Table 5 and Figure 4).

Table 7. Correlation table

		Anger Scale	Satisfaction With Life Scale	Ageism Scale	Expectancy effect size
Anger Scale	r	1	.009	.069	-.044
	p		.954	.672	.788
	n	40	40	40	40
Satisfaction With Life Scale	r	.009	1	.140	-.073
	p	.954		.388	.656
	n	40	40	40	40

		Anger Scale	Satisfaction With Life Scale	Ageism Scale	Expectancy effect size
Ageism Scale	r	.069	.140	1	-.132
	p	.672	.388		.415
	n	40	40	40	40
Expectancy effect size	r	-.044	-.073	-.132	1
	p	.788	.656	.415	
	n	40	40	40	40

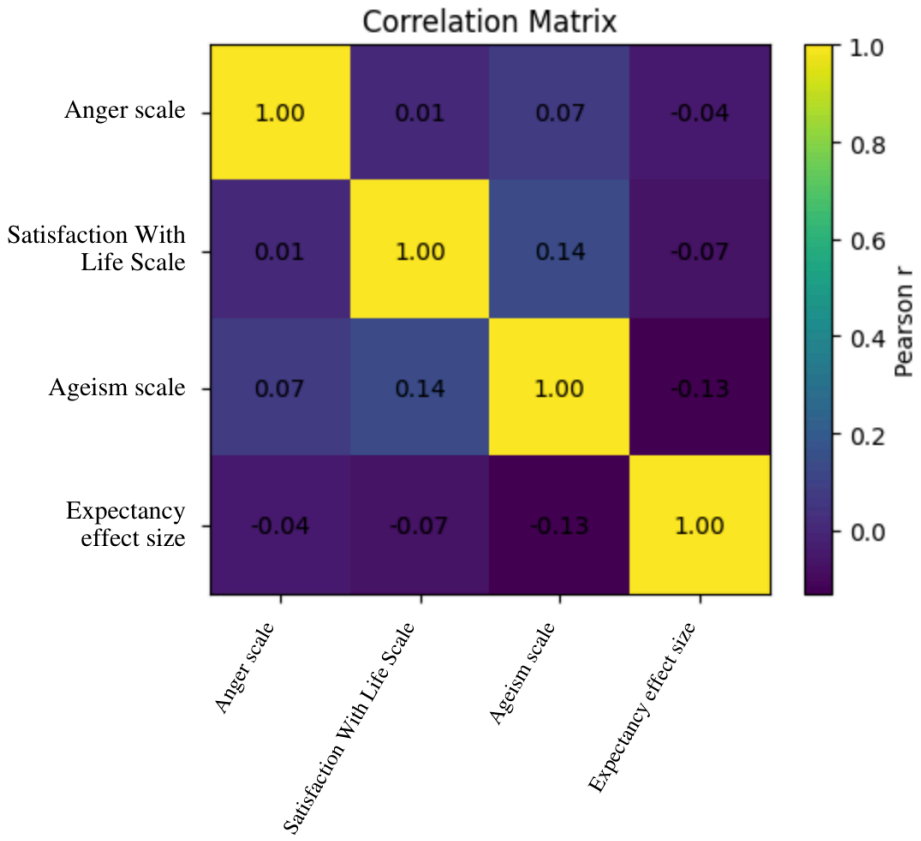


Figure 4. Correlation matrix

5. Discussion

The aim of this bachelor's thesis was to examine how time-based expectancy affects performance on a facial gender categorization task, and the extent to which this process may

change under the influence of socially significant factors such as one's own gender, stimulus gender, age, and emotional expression. The study used a modified version of the time–event correlation paradigm, in which a face of a particular sex was associated with a particular foreperiod, a short or long interval, in most trials.

The main finding was that participants responded faster to stimuli presented in predictable combinations than to stimuli presented in unpredictable combinations. The mixed ANOVA of reaction time revealed a main effect of expectancy, indicating that time-based expectancy formed during performance of the gender categorization task. Thus, the first hypothesis was supported: temporal expectancy did reduce reaction time and facilitate response preparation when the timing of stimulus onset and the stimulus category were predictable. This result is consistent with the existing literature on time-based expectancy, according to which responses are more efficient when a particular type of event is expected at a particular time (Thomaschke & Dreisbach, 2013). Furthermore, our study established for the first time that time-based event expectancy forms during a gender categorization task, indicating that it is possible to form time-based expectancy for a social stimulus, specifically its gender characteristics.

The interaction among expectancy, stimulus sex, and participant group was particularly important. Additional t-test analyses showed that, among female participants, the temporal expectancy effect emerged for female face stimuli, whereas the same effect was not observed for male faces. Among male participants, no significant temporal expectancy effect was found for either male or female faces. Thus, the second hypothesis was partially supported: female participants showed a temporal expectancy effect for own-gender stimuli, but an analogous tendency was not observed in male participants.

This finding is consistent with research indicating that facial gender is a socially significant category and may influence the behavioral and neural processing of faces. Female participants have been shown to tend to recognize female faces better, while stronger activity during the processing of female faces has been observed in areas associated with face processing, including the fusiform gyrus and inferior occipital gyrus (Lovén et al., 2014). This suggests that

faces of one's own gender group may be processed more efficiently at both early visual and higher configural levels.

Women's own-gender bias may be based not only on holistic processing of the whole face, but also on more detailed representations of individual facial parts (Hausinger et al., 2025). If female faces are processed in greater detail or given greater priority by female participants, it is possible that temporal regularities associated with these stimuli are also formed more readily. Accordingly, the expectancy effect observed for female faces among female participants in this study may be considered a manifestation of own-gender bias in temporal–event prediction.

The results are also related to the general approach to social categorization. When processing social information, people often use categories such as sex, age, or group membership to make person perception faster and more efficient (Macrae & Bodenhausen, 2000). In a gender categorization task, participants process not only a face's physical characteristics, but also a social category. Therefore, the temporal expectancy effect in this study may reflect the integration of temporal prediction with social categorization. In other words, participants learned not only when a stimulus would appear, but also that a face of a particular sex was more likely to appear at a particular time.

The third hypothesis, according to which emotional and age-related social factors would interact with time-based expectancy, was not supported. The main effects of emotional expression and age, as well as their interactions with expectancy, were not statistically significant.

Overall, the study showed that time-based expectancy influences performance on a facial gender categorization task. Participants responded faster to predictable stimuli than to unpredictable stimuli, confirming that temporal regularities can be used to process socially significant visual stimuli. At the same time, the temporal expectancy effect was not entirely general or uniform: it was particularly evident for female faces among female participants. This finding indicates that the formation of temporal expectancy may be influenced not only by when a stimulus appears, but also by its social-categorical significance. However, the influence of emotional expression and age was not confirmed, suggesting that temporal expectancy in this paradigm was more closely associated with gender category than with other social cues.

5.1. Limitations and Directions for Future Research

Several limitations should be considered when interpreting the results of this study. First, the third hypothesis, according to which emotional expression, age, and time-based expectancy would interact with one another, was not supported. One possible reason may be the relatively small sample size. Forty participants took part, which was sufficient to detect the main effect of temporal expectancy and the gender-related interaction, but may have been insufficient to detect more complex interactions. Increasing the number of participants will be important in future studies.

A second important limitation concerns the participants' age group. Only young participants took part in this study, owing to limited resources. Although the stimuli included both young and older faces, the participant group itself was homogeneous in age. This limits a full assessment of age-related effects. Adding a group of older participants in a future study would make it possible to test whether temporal expectancy forms differently for faces from one's own and other age groups. Such a design may better reveal the role of age-related social categorization in a gender categorization task.

In addition, the gender factor may have been so strong and dominant in this paradigm that it obscured possible effects of emotion and age. It would therefore be desirable to use an experimental design in which the interaction between gender and emotion is tested separately, without the age factor or under strict control of age. For example, faces from only one age category could be used to assess whether emotional expression influences the formation of temporal expectancy during gender categorization. The interaction between age and gender could then be examined separately while controlling emotional expression.

6. Conclusions

- The study found that temporal expectancy formed during performance of the gender categorization task.

- A gender bias effect was found among female participants: time-based expectancy formed more efficiently for female face stimuli than for male face stimuli. This tendency was not observed among male participants.
- During performance of the gender categorization task, the age and emotional expression of the facial stimulus did not modulate the formation of time-based expectancy.

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Appendices

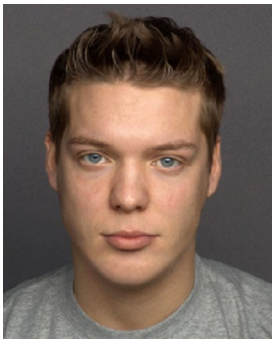
Appendix 1: Consent Form

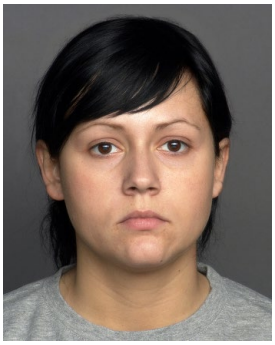
Brief description of the experiment

We invite you to participate in an experiment aimed at studying face perception. The tasks you will perform resemble computer games. We will use various questionnaires to assess your personal attitudes.

Task 1

Faces will appear in the center of the computer screen. The participant's task is to press the right-hand key when a female face appears and the left-hand key when a male face appears, as quickly and accurately as possible.





The maximum duration of the experiment is approximately 40–50 minutes.

The observation itself involves no risk. Participation is voluntary, and participants have the right to discontinue their participation at any time. The study will be conducted in full accordance with the principles of confidentiality. Your first and last names will not be used in any document.

By signing, you agree to participate in this study.

Participant's signature _____ Date _____

Participant's first/last name _____ _____

Telephone number _____

If you have any questions about the study, please contact us:

Appendix 2: General Information

General Information

Participant code: _____ Date: _____

Sex: _____ Age: _____ Date of birth: _____

Total number of years of education _____

Academic level _____

Native language _____

Other languages _____

Left-handed ☐ Right-handed ☐

Are you taking any medication?

No ☐

Yes ☐

Please specify which medication and for how long.

_____

Do you see a doctor regularly?

No ☐

Yes ☐

Please specify the type and duration of the problem.

Have you ever had convulsions/seizures/epilepsy?

No ☐

Yes ☐

Please specify how often and, if you received treatment, which medications were used.

_____

Have you ever had a serious head injury that caused a loss of consciousness or hospitalization?

No ☐

Yes ☐

Please specify when the injury occurred.

Appendix 3: Satisfaction With Life Scale

Satisfaction With Life Scale

ID: _____ Date: _____

Please select one response for each statement:

1. In most ways, my life is close to my ideal.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐

2. The conditions of my life are excellent.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐

3. I am satisfied with my life.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐

4. So far, I have achieved all the important things I wanted in life.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐

5. If I could start my life over, I would change almost nothing.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐

Scale

1 = Strongly disagree

2 = Disagree

3 = Somewhat disagree

4 = Neither agree nor disagree

5 = Somewhat agree

6 = Agree

7 = Strongly agree

Appendix 4: Anger Scale of the Aggression Questionnaire

Emotional Reactivity/Irritability Scale

ID: _____ Date: _____

Please select one response for each statement:

1. I get angry easily, but the feeling soon passes.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

2. When I am frustrated, I show my irritation.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

3. Sometimes I feel like a powder keg ready to explode.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

4. I am an even-tempered person.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

5. Some of my friends think I am hot-tempered.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

6. Sometimes I lose my temper for no reason.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

7. I have difficulty controlling my temper.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

Scale

1 – Completely uncharacteristic

2 – Mostly uncharacteristic

3 – Neither characteristic nor uncharacteristic

4 – Mostly characteristic

5 – Completely characteristic

Appendix 5: Questionnaire for Assessing Negative Stereotypes About Older People

Questionnaire on Aging

ID: _____ Date: _____

Please select one response for each statement:

a) Strongly agree b) Agree c) Disagree d) Strongly disagree

1. Most adults maintain a satisfactory level of health until approximately age 65, after which their health deteriorates sharply.

a ☐ b ☐ c ☐ d ☐

2. Most people over 65 have a range of limitations that make them dependent on others.

a ☐ b ☐ c ☐ d ☐

3. Most people begin to experience a significant decline in memory at around age 65.

a ☐ b ☐ c ☐ d ☐

4. Learning something new is almost impossible for older people.

a ☐ b ☐ c ☐ d ☐

5. As they age, older people become increasingly rigid and inflexible.

a ☐ b ☐ c ☐ d ☐

6. As people grow older, they become less active.

a ☐ b ☐ c ☐ d ☐

7. In general, older people are less responsible than young people.

a ☐ b ☐ c ☐ d ☐

8. It is preferable for people to retire at the age of 60–65.

a ☐ b ☐ c ☐ d ☐

9. Older people have few friends.

a ☐ b ☐ c ☐ d ☐

10. Older people are less interested in sex.

a ☐ b ☐ c ☐ d ☐

11. Since work is important to people, retirement is likely to cause serious psychological distress.

a ☐ b ☐ c ☐ d ☐

12. As we grow older, we lose the ability to solve the problems we face.

a ☐ b ☐ c ☐ d ☐

13. Older people are often like children.

a ☐ b ☐ c ☐ d ☐

14. As we age, we lose interest in various activities.

a ☐ b ☐ c ☐ d ☐

15. People's shortcomings become more pronounced with age.

a ☐ b ☐ c ☐ d ☐

16. Older people become irritated easily and are "grumpy."

a ☐ b ☐ c ☐ d ☐

17. Old age is a time to rest after a lifetime of work.

a ☐ b ☐ c ☐ d ☐

18. Physical exercise is not recommended for older people.

a ☐ b ☐ c ☐ d ☐

19. Almost no one over 65 performs a job as well as a young person.

a ☐ b ☐ c ☐ d ☐

20. A substantial proportion of people over 65 are "old-fashioned thinkers."

a ☐ b ☐ c ☐ d ☐